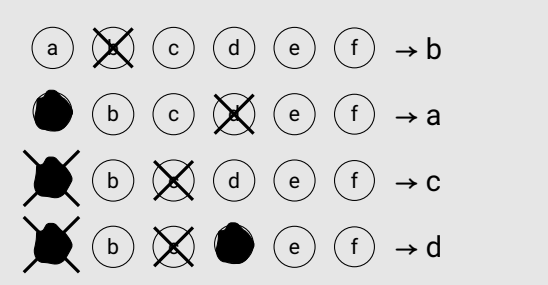
Exercises

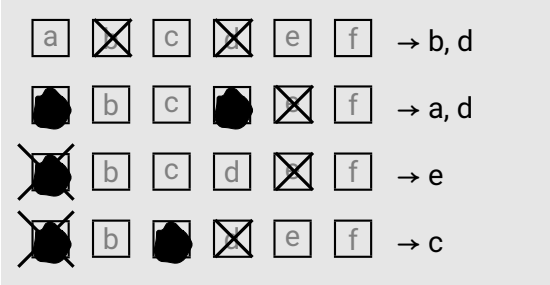
1	2	3	4	5	6	7

Surname, First name

1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9
0	0	0	0	0	0	0	0

Machine Learning 1 (52041MAL6Y)
Final exam
24 Oct, 2025, 09:00-12:00





Answer multiple-choice questions as shown in the example. Circular checkboxes have only one correct answer. Square checkboxes may have multiple correct answers.

Pay attention to the following;

- **WRITE YOUR NAME AND STUDENT NUMBER ON THE FRONT PAGE** (don't forget to mark the digits)
- Write all your answers on this exam booklet; it will be scanned and digitally graded.
- Write your **answers inside the boxes** (it is ok, to slightly go over the margins though).
- The answer boxes should be large enough for your answer
- **If you need to empty the answer box in order to start over, ask the invigilator for a blank sticker or use the *Emergency Box* at the end of the booklet.**

During the exam you are allowed to use:

- The provided cheat sheet.
- Basic calculator (although you will not need it)

Please scan over the questions before you start to get an impression of the content

- You have 3 **hours** for the exam (except for those with pre-approved extensions)
- In total **40 points** can be earned divided over 6 exercises:
  - Exercise 1: Multiple choice questions [9p]
  - Exercise 2: True/False statements [6p]
  - Exercise 3: Neural networks [4p]
  - Exercise 4: Dimensionality reduction [4p]
  - Exercise 5: Mixture of Experts in the *Mushroom Kingdom* [6p]
  - Exercise 6: L2 Support Vector Machine [11p]
- We work with a partial grading: you can get points even if you don't manage to solve the full question.

[9p] Multiple choice questions

*NOTE: Questions 1a-c have **multiple** correct options (you need to mark all correct options to get full score). In questions 1d-g and 3a there is **only one** correct answer.*

1.5p **1a** Which of the following are generative models?

- ☐ Linear regression.
- ☐ Logistic regression.
- ☐ Multi-layer perceptrons.
- ☐ K-Means.
- ☐ Gaussian Mixture Models.
- ☐ Support Vector Machines.

1.5p **1b** Which of the following statements about Probabilistic PCA (PPCA) are true?

- ☐ The relationship between latent and observed variables in PPCA is nonlinear.
- ☐ The latent variable in PPCA is assumed to follow a standard Gaussian distribution.
- ☐ When the noise variance approaches zero, PPCA yields the same principal components as PCA.
- ☐ PPCA is a generative model.

2p **1c** Let $\mathcal{D} = \{(x_i, t_i)\}_{i=1}^N$ with samples given by $t_i = f(x_i)$ with $f \sim GP(m(\cdot), k(\cdot,\cdot))$ a random function according to a Gaussian process with mean function m and kernel k. Which statements are true?

- ☐ t_i is not a random variable.
- ☐ $t_i \sim p(t)$ is i.i.d. relative to some $p(t)$.
- ☐ $t_i \sim p(t|x_i)$ is i.i.d. relative to some $p(t|x)$
- ☐ $\mathrm{Cov}[t_i, t_j] = 0$ for any $i \neq j$
- ☐ $\mathrm{Cov}[t_i, t_j] = k(t_i, t_j)$
- ☐ $\mathrm{Cov}[t_i, t_j] = k(x_i, x_j)$

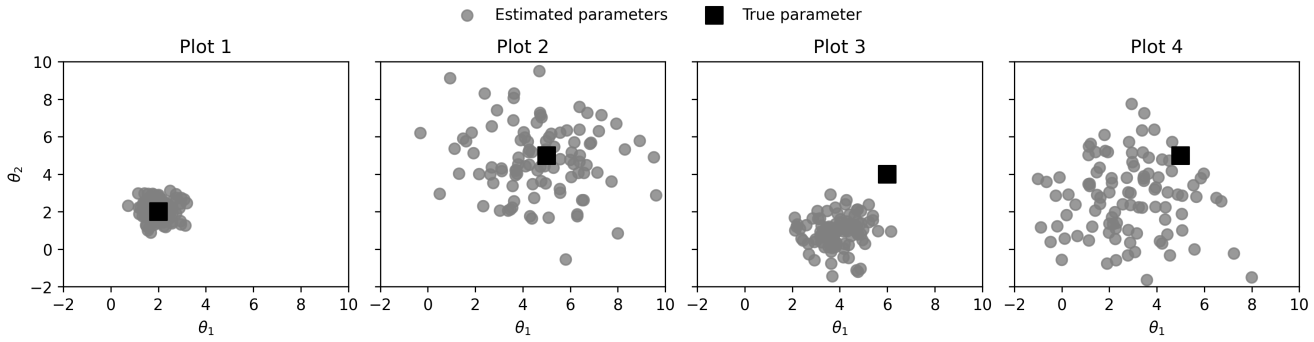


Figure 1: Parameter estimates for different models. Relevant for question 1d.

1p **1d** We consider 4 different models. We estimate each model's parameter vector $\boldsymbol{\theta} = [\theta_1, \theta_2]^T$ using 50 different training datasets. The results are plotted in Figure 1.

For each plot, specify whether the corresponding model has high/low bias and high/low variance.

(a) (1) low bias, low variance (2) low bias, high variance (3) high bias, low variance (4) high bias, low variance.

(b) (1) low bias, low variance (2) high bias, low variance (3) low bias, high variance (4) high bias, high variance.

(c) (1) low bias, low variance (2) low bias, high variance (3) high bias, low variance (4) high bias, high variance.

(d) (1) high bias, high variance (2) low bias, low variance (3) low bias, high variance (4) low bias, low variance.

1p **1e** Which of the following kernels implicitly maps data into an infinite-dimensional feature space?

(a) $k(\mathbf{x}, \mathbf{y}) = \mathbf{x}^\top \mathbf{y}$.

(b) $k(\mathbf{x}, \mathbf{y}) = (\mathbf{x}^\top \mathbf{y} + c)^d, \quad c \geq 0.$

(c) $k(\mathbf{x}, \mathbf{y}) = \exp\left(- \frac{\|\mathbf{x}-\mathbf{y}\|^2}{2\sigma^2} \right).$

(d) $k(\mathbf{x}, \mathbf{y}) = \frac{\mathbf{x}^\top \mathbf{y}}{\|\mathbf{x}\| \, \|\mathbf{y}\|}.$

1p **1f** We consider the dataset $\mathbf{X} = \{\mathbf{x}_n\}_{n=1}^{N}$, where $\mathbf{x}_n \in \mathbb{R}^5$. We carry out an eigendecomposition of the empirical covariance matrix $\mathbf{S}$, obtaining $\mathbf{S} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^{\mathbf{T}}$, where

$$\mathbf{\Lambda} = \begin{pmatrix} 12 & 0 & 0 & 0 & 0 \\ 0 & 12 & 0 & 0 & 0 \\ 0 & 0 & 8 & 0 & 0 \\ 0 & 0 & 0 & 6 & 0 \\ 0 & 0 & 0 & 0 & 2 \end{pmatrix}.$$

We apply Principal Component Analysis to reduce the dimensionality of $\mathbf{X}$. What is the minimum dimensionality of the reduced dataset such that at least 75% of the variance is preserved?

(a) 1 (b) 2 (c) 3 (d) 4 (e) 5

1p **1g** Consider the following constrained optimization problem:

$$\max_{\mathbf{x} \in \mathbb{R}^2} f(\mathbf{x}) \quad \text{subject to} \quad g(\mathbf{x}) \leq 0.$$

Let $\mathbf{x}^*$ be a solution located in the boundary ($g(\mathbf{x}^*) = 0$). What can we say about $\nabla_{\mathbf{x}} f(\mathbf{x}^*)$ and $\nabla_{\mathbf{x}} g(\mathbf{x}^*)$?

(a) They are parallel ($\nabla_{\mathbf{x}} f(\mathbf{x}^*) = \mu \nabla_{\mathbf{x}} g(\mathbf{x}^*),\ \mu \geq 0$).

(b) They are anti-paralell ($\nabla_{\mathbf{x}} f(\mathbf{x}^*) = -\mu \nabla_{\mathbf{x}} g(\mathbf{x}^*),\ \mu \geq 0$).

(c) They are equal.

(d) They are both zero.

[6p] True/False statements

Determine whether the following statements are True or False. **Justify your answer** by giving a proof/explanation or a counterexample. Note that you will only get points if your explanation is correct.

1.5p **2a** We define the reReLU activation function as

$$\text{reReLU}(x) = \text{ReLU}(\text{ReLU}(x)).$$

A neural network that uses reReLU as activation function in all hidden layers is able to learn more complex non-linearities than one that uses ReLU in all hidden layers.

Circle your answer: True / False

1.5p **2b** *The function* $k(\mathbf{x}, \mathbf{x}') = \mathbf{x}^\top \mathbf{x}' + 2\mathbf{x}^\top \mathbf{x}$ *is a valid kernel.*

Circle your answer: True / False

1.5p **2c** We consider the following SVM optimization problem:

$$\min_{\mathbf{w},b,\xi} \frac{1}{2}\|\mathbf{w}\|^2 + \frac{1}{\lambda}\sum_{n=1}^{N}\xi_n \text{ s.t } \begin{cases} t_n\left(\mathbf{w}^{\mathbf{T}}\mathbf{x_n} + b\right) & \geq 1-\xi_n \quad \forall n \in \{1,2,...,N\}, \\ \xi_n & \geq 0 \quad \forall n \in \{1,2,...,N\}. \end{cases}$$

For a large value of λ, there will be less support vectors.

Circle your answer: True / False

1.5p **2d** We consider a two-dimensional dataset with data points belonging to two different classes. The dataset is projected separately onto its first principal component (resulting in Dataset 1) and its second principal component (resulting in Dataset 2). We then train a linear SVM with the same regularization parameter on each of these projections to classify the two classes.

The SVM trained on Dataset 1 will necessarily achieve higher classification accuracy than the SVM trained on Dataset 2.

Circle your answer: True / False

[4p] Neural networks

Consider the following Multilayer Perceptron (MLP) architecture:

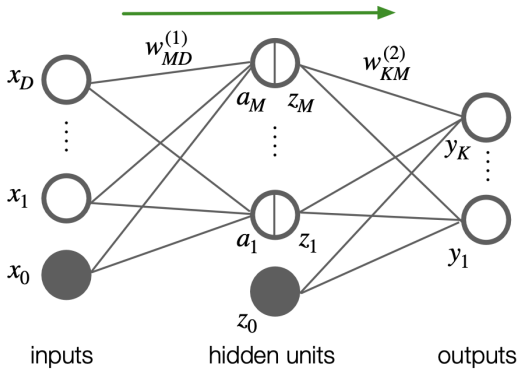


Figure 2: Two-layer MLP architecture. Relevant for questions 3a-3d. Note that x_0 and z_0 are bias units, which are always equal to 1.

1p **3a** How many trainable parameters does the model from Figure 2 have?

(a) $DM + MK$

(b) $D^2 + K^2$

(c) $D(D+1) + K(K+1)$

(d) $M(D+1) + K(M+1)$

(e) $D(M+1) + M(K+1)$

1.5p **3b** We want to use the architecture sketched in Figure 2 for a **classification** task, where we receive an input vector $\mathbf{x} \in \mathbb{R}^D$ and want to classify it into one out of K different classes. We have a dataset of N samples with their corresponding ground-truth labels.

1. What would we need to modify or add to our current architecture to turn it into a classification model that outputs a class label?
2. What loss function should we optimize?

*Note: we are expecting a **brief list of necessary layers/functions**, indicating the **name of the function** and what **its input** would be. You do not need to write any formulas.*

1.5p **3c** We want to use the architecture sketched in Figure 2 for a **regression** task, where we receive an input vector $\mathbf{x} \in \mathbb{R}^D$ and want to predict a single scalar target $t \in \mathbb{R}$. We have a dataset of N samples with their corresponding ground-truth target values.

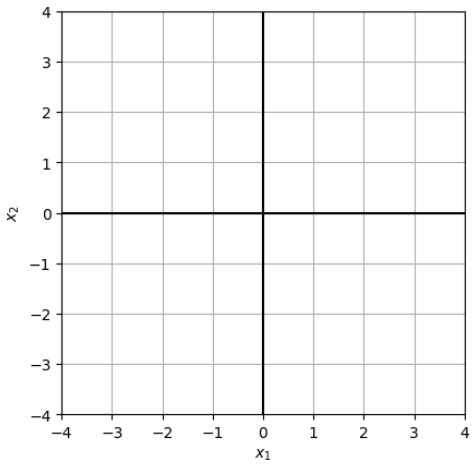
1. What would we need to modify or add to our current architecture to turn it into a regression model that outputs a scalar target?
2. What loss function should we optimize?

*Note: we are expecting a **brief list of necessary layers/functions**, indicating the **name of the function** and what **its input** would be. You do not need to write any formulas.*

[4p] Dimensionality reduction

We apply Principal Component Analysis (PCA) to a 2-dimensional dataset $\mathbf{X}$, obtaining that its first principal component is $\mathbf{u}_1 = \left(\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}\right)^T$ and its second principal component is $\mathbf{u}_2 = \left(-\frac{2}{\sqrt{5}}, \frac{1}{\sqrt{5}}\right)^T$.

1p **4a** Draw an example of a dataset $\mathbf{X}$ that satisfies this on the plot below. Include at least 10 samples.



For questions 4b and 4c we will assume that $\mathbf{X}$ is a dataset with zero mean.

0.5p **4b** Compute the projection of $\mathbf{x}_0 = (2, 2)$ onto the subspace defined by the first principal component.

1p **4c** If we represent the projection of $\mathbf{x}_0 = (2, 2)$ on the first component in the original 2D space, what will be the reconstruction error with respect to the original datapoint?

1.5p **4d** In this course, we have also introduced autoencoders for dimensionality reduction. Briefly explain how they work. Specifically:

1. Draw a sketch of the architecture, indicating the dimensionality of each layer.
2. Write down the optimization objective.
3. How can standard PCA be formulated as a special case of an autoencoder?

[6p] Mixture of Experts in the Mushroom Kingdom

Mamma mia! The evil *Bowser* has kidnapped *Princess Peach* again. *Mario* is about to start his journey to *Bowser*'s castle, but, this time, he wants to apply some of the knowledge he acquired during the ML1 course to make his path a bit easier. More specifically, he wants to model the time between *piranha plants* coming out of their pipes to gain an advantage over them.

Mario has found some old files in *Princess Peach*'s castle that state the following:

- There are three types of piranha plants: *regular*, *fire*, and *ice*. For the sake of convenience you can refer to these types with respectively class lables C_1, C_2, and C_3.
- For a specific type of piranha plant, the time between pop-outs follows an exponential distribution, which has the following density:

$$p(x \mid \lambda) = \lambda e^{-\lambda x}, \quad x \geq 0, \lambda > 0.$$

He also found a data file with a list of pop-up times of piranha plants, but the plant types are not specified.

2p **5a** Given the information he gathered, *Mario* decided to model the time between appearances of piranha plants of any type with a mixture of exponential distributions. Do/answer the following:

- Write down the probability density function for such model.
- Clearly define each of the random variables involved.
- How many trainable parameters does the model have?

1p **5b** Write the log-likelihood function for a given dataset of samples $D = \{x_1, x_2, \ldots, x_N\}$ for this model.

1p **5c** Write down the expression for the responsibility of *expert* k on datapoint n, i.e., $r_{nk} = p(z = C_k \mid x_n)$.

1p **5d** *Mario* now wants to apply the Expectation-Maximization algorithm to learn optimal parameters for the model. Formulate the optimization problems that need to be solved to update the parameters in the Maximization step.

Note: You do not need to compute the analytical expressions for the parameter updates, but ***only specify the optimization problem that needs to be solved to obtain them.***

1p **5e** With his model trained, *Mario* enters a completely dark cave where he can't see anything. Suddenly, he hears a *piranha plant* nearby. By listening closely, he notices it pops out from its pipe every 5 seconds. How can he use his trained model to predict which type of *piranha plant* it is?

[11p] L2 Support Vector Machine

Let $\{(\mathbf{x}_n, t_n)\}_{n=1}^N$ be a set of N training pairs of feature vectors $\mathbf{x}_n$ and labels $t_n \in \{-1, +1\}$. The following is the primal formulation of L2-SVM, a variant of the standard SVM obtained by squaring the penalty for the slack variables:

$$\begin{aligned} \min_{\mathbf{w},b,\boldsymbol{\xi}} \quad & \tfrac{1}{2}\mathbf{w}^\top\mathbf{w} + \frac{C}{2}\sum_{n=1}^{N}\xi_n^2 \\ \text{s.t.} \quad & t_n(\mathbf{w}^\top\mathbf{x}_n + b) \geq 1 - \xi_n, \quad \forall n \in \{1, ..., N\}, \\ & \xi_n \geq 0, \quad \forall n \in \{1, ..., N\}. \end{aligned}$$

Note: Question a) is difficult, but it is not needed to solve the rest of the subquestions.

2p **6a** Show that removing the constraints $\xi_n \geq 0$ does not change the optimal solution to the primal problem (i.e., proof that the constraints $\xi_n \geq 0$ are no longer necessary in this L2-SVM formulation).

Hint: assume there is an optimal solution such that $\xi_j^* < 0$ for some j and try reach a contradiction.

After removing the constraints for the slack variables, we obtain a relaxed optimization problem, which is the one that we will consider in the rest of subquestions.

$$\begin{aligned} \min_{\mathbf{w},b,\boldsymbol{\xi}} \quad & \tfrac{1}{2}\mathbf{w}^\top\mathbf{w} + \tfrac{C}{2}\sum_{n=1}^{N}\xi_n^2 \\ \text{s.t.} \quad & y_n(\mathbf{w}^\top\mathbf{x}_n + b) \geq 1 - \xi_n, \quad \forall n \in \{1, ..., N\}. \end{aligned}$$

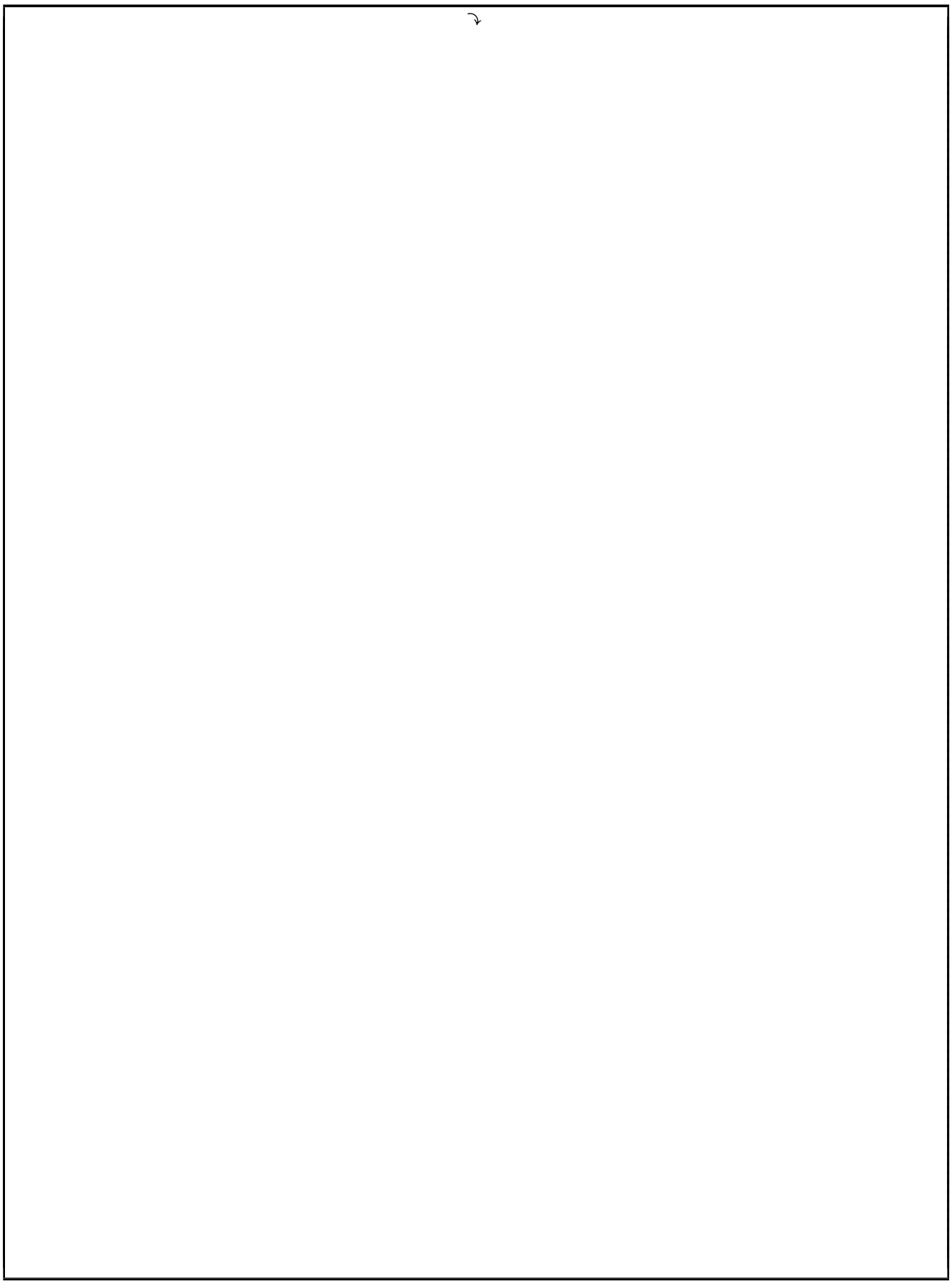
0.75p **6b** Write the primal Lagrangian for the relaxed optimization problem. Denote the Lagrange multipliers as $\{\alpha_n\}_{n=1}^N$.

1p **6c** List the KKT conditions. How many (scalar) conditions are there?

1.5p **6d** Derive the stationary conditions for the primal Lagrangian.

1p **6e** Provide the definition of the dual Lagrangian (i.e., how is it defined in terms of the primal Lagrangian?) and formulate the dual optimization problem.

2.5p **6f** Derive the dual Lagrangian.



1.5p **6g** We obtain an optimal solution $\boldsymbol{\alpha}^*$ for the dual problem. Compute the optimal primal variables $\mathbf{w}^*$, b^* and $\boldsymbol{\xi}^*$ in terms of $\boldsymbol{\alpha}^*$.

0.75p **6h** How can we use the optimal parameters $(\mathbf{w}^*, b^*)$ to classify a test datapoint $\mathbf{x}$?

Emergency box

7 *Use this space only as a last resort* to provide answer to a previous question that didn't fit in the corresponding box. Indicate which question it is about (and also indicate it in the answer box of the corresponding question).